

Case Study: The IT Act 2000 and the 2019 SBI Data Breach

Introduction

The Information Technology Act, 2000 (IT Act) was enacted in India to provide a legal framework for electronic governance, cybercrime regulation, and data protection. This case study explores the 2019 data breach involving the State Bank of India (SBI), analyzing the implementation of the attack, how the attackers were caught, the general punishment for such crimes, and the specific outcomes of this case.

Attack Overview

Implementation of the Attack

In 2019, a significant data breach occurred involving the State Bank of India (SBI), one of India's largest public sector banks. The breach was primarily executed through a phishing attack, which exploited vulnerabilities in the bank's cybersecurity defenses.

1. **Phishing Scheme:** Attackers sent out fake emails designed to look like they were from SBI. These emails contained malicious links that, when clicked, redirected users to a counterfeit SBI website.
2. **Data Harvesting:** On the fraudulent site, users were prompted to enter their personal banking details, including login credentials and transaction PINs. The attackers collected this sensitive information as users entered it.
3. **Exploitation:** With the harvested data, the attackers gained unauthorized access to multiple accounts. They executed fraudulent transactions, siphoning off funds and compromising sensitive personal information of numerous customers.

Detection and Capture

How the Attackers Were Caught

The breach was detected through several mechanisms:

1. **Anomalous Activity:** Suspicious patterns were noticed by SBI's cybersecurity team, including unusual login attempts and unauthorized transactions.
2. **Customer Complaints:** Numerous customers reported unauthorized transactions on their accounts, prompting a closer investigation by the bank.
3. **Investigation:** The investigation revealed that the phishing sites were registered using fake identities. By tracing the IP addresses and domain registration details, law enforcement agencies, in collaboration with cybersecurity experts, were able to identify

the perpetrators.

4. **Law Enforcement:** The case was taken up by the Cyber Crime Cell of the local police. After an extensive investigation involving digital forensics and coordination with international law enforcement, the primary suspects were apprehended.
5. **Arrests:** The attackers were arrested in multiple locations across India. Evidence gathered included logs of phishing emails, fraudulent domain registrations, and communication records linking the suspects to the crime.

Legal Framework and Punishment

General Punishment for Cyber Crimes

Under the IT Act 2000, cyber crimes such as data theft, phishing, and unauthorized access are punishable by law. The relevant sections include:

1. **Section 66D:** For punishment related to cheating by personation using computer resources, which can result in imprisonment up to three years or a fine up to one lakh rupees, or both.
2. **Section 66E:** Deals with the violation of privacy, including capturing or transmitting images of individuals without their consent, punishable by imprisonment up to three years or a fine up to two lakh rupees, or both.
3. **Section 43:** Pertains to damage to computer systems, and the penalty can include compensation for damages suffered by the victim.

Specific Outcome in the SBI Case

In the SBI data breach case:

1. **Charges:** The attackers were charged under various sections of the IT Act, including Section 66D for phishing and Section 43 for unauthorized access and damage.
2. **Punishment:** The court imposed a combination of imprisonment and fines on the perpetrators. The primary individuals involved received sentences ranging from two to four years, along with substantial monetary fines as compensation to the affected customers and the bank.
3. **Restitution:** Additionally, the court mandated that the convicted individuals provide restitution to the victims to compensate for their financial losses.

Execution of the Attack

Execution Method

The attack was meticulously planned and executed as follows:

1. **Social Engineering:** The attackers used social engineering tactics to deceive users into divulging sensitive information.
2. **Technical Execution:** By creating a convincing fake website, they managed to capture login credentials and transaction PINs.
3. **Data Manipulation:** Using the stolen credentials, the attackers accessed and manipulated bank accounts, initiating unauthorized transactions to siphon funds.
4. **Cover-Up:** The attackers employed various techniques to hide their tracks, including using VPNs and anonymizing services to mask their IP addresses and geographic locations.

Conclusion

The 2019 SBI data breach underscores the evolving nature of cyber threats and the critical importance of robust cybersecurity measures. The IT Act 2000 provides a comprehensive legal framework for addressing such crimes, though the rapid pace of technological advancement necessitates continuous updates to legal and regulatory standards. The case also highlights the importance of proactive cybersecurity practices and swift law enforcement responses in mitigating the impact of cybercrime.